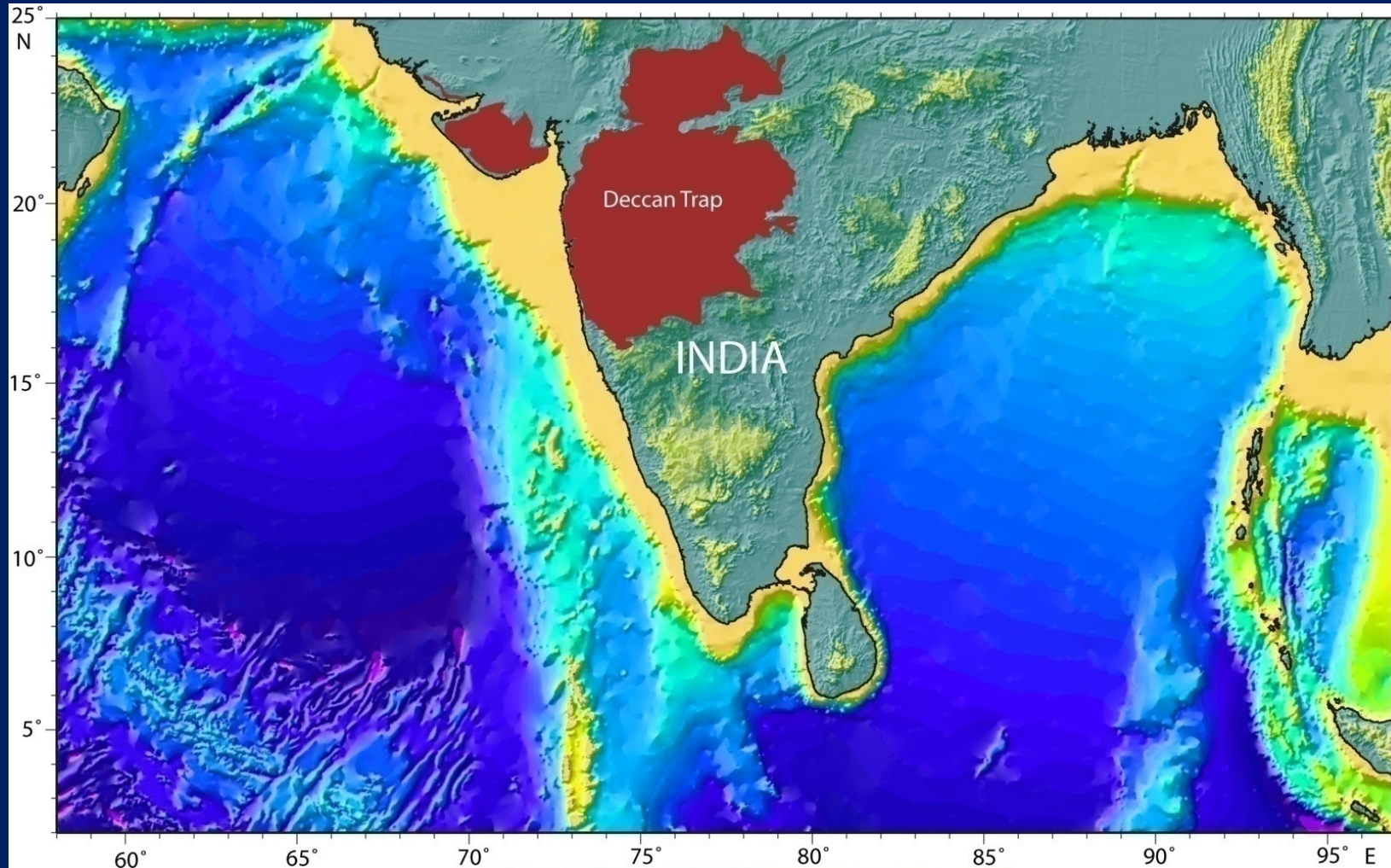


# Carbon Reservoir and Fluxes



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# The Carbon : Significance

- Carbon is the foundation of all life on Earth, required to form complex molecules like proteins and DNA.
- The carbon is also found in our atmosphere in the form of carbon dioxide (CO<sub>2</sub>).
- Carbon helps to regulate the Earth's temperature, makes all life possible, is a key ingredient in the food that sustains us, and provides a major source of the energy to fuel our global economy.

# Basic concepts

- **Carbon Reservoirs** - forms in which carbon resides within the earth system- usually expressed in terms of the mass of carbon in Gigatons (Gt) = Petagram (Pg)= $10^{15}$ g
- **Carbon Transfer Mechanisms** - processes that move carbon between reservoirs - they usually involve a physical process and a chemical reaction.
- **Carbon Transfer Rate** - expressed in terms of Pg per year
- **Carbon Residence Time** – residence time for carbon in a reservoir is estimated by dividing the amount of carbon in that reservoir by the transfer rates in and out of it.

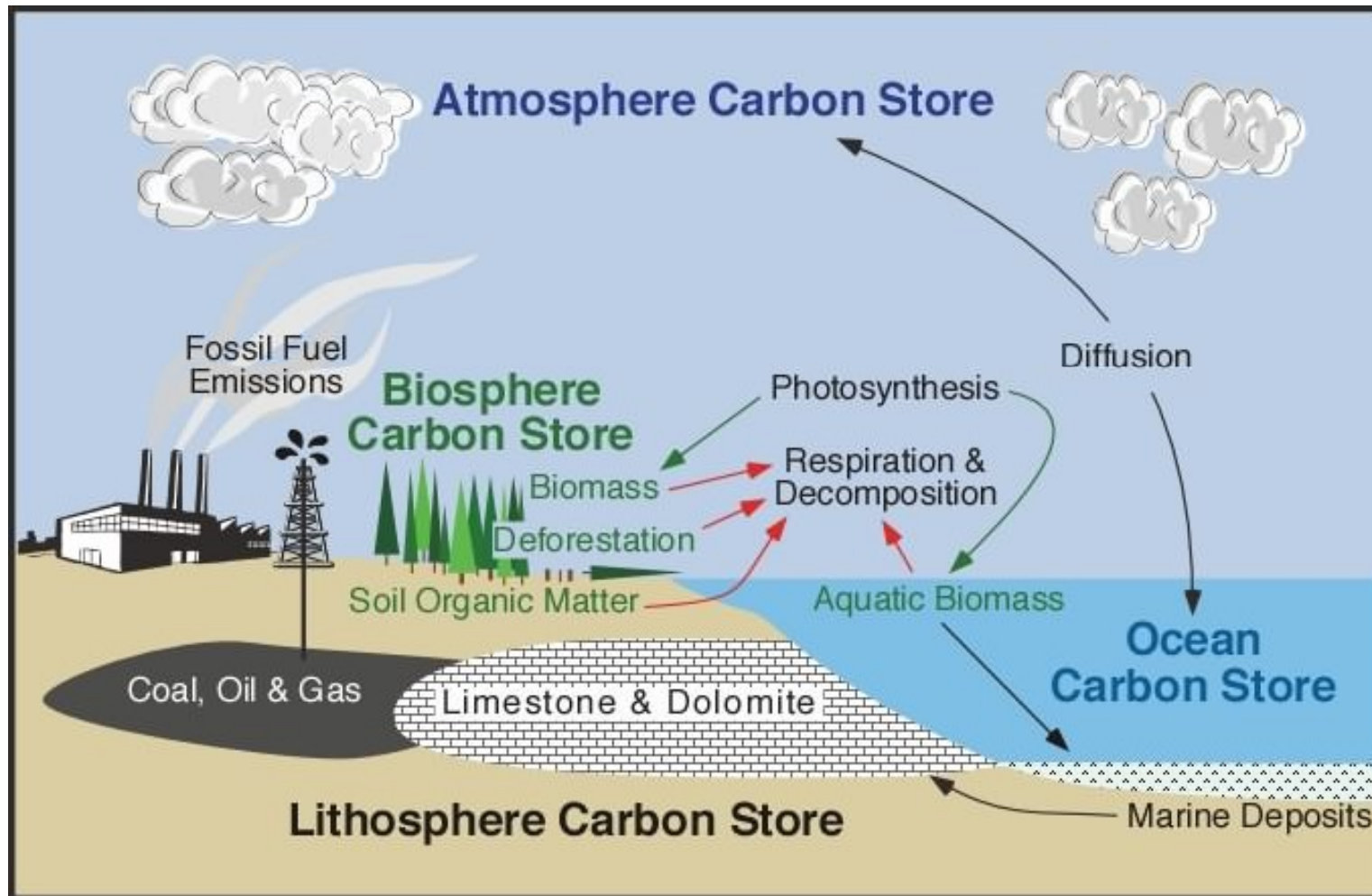
# Carbon reservoirs

- The atmosphere
- The biosphere (Terrestrial ecosystems) (include fresh water systems, non-living organic material, such as soil carbon and organic forms like animals and plants.
- The oceans (including dissolved inorganic carbon and living and non-living marine biota).

## Carbon reservoirs.....contd.

- The lithosphere (sediments, Earth's crust including fossil fuels).
- The Cryosphere the frozen water part of the Earth system.

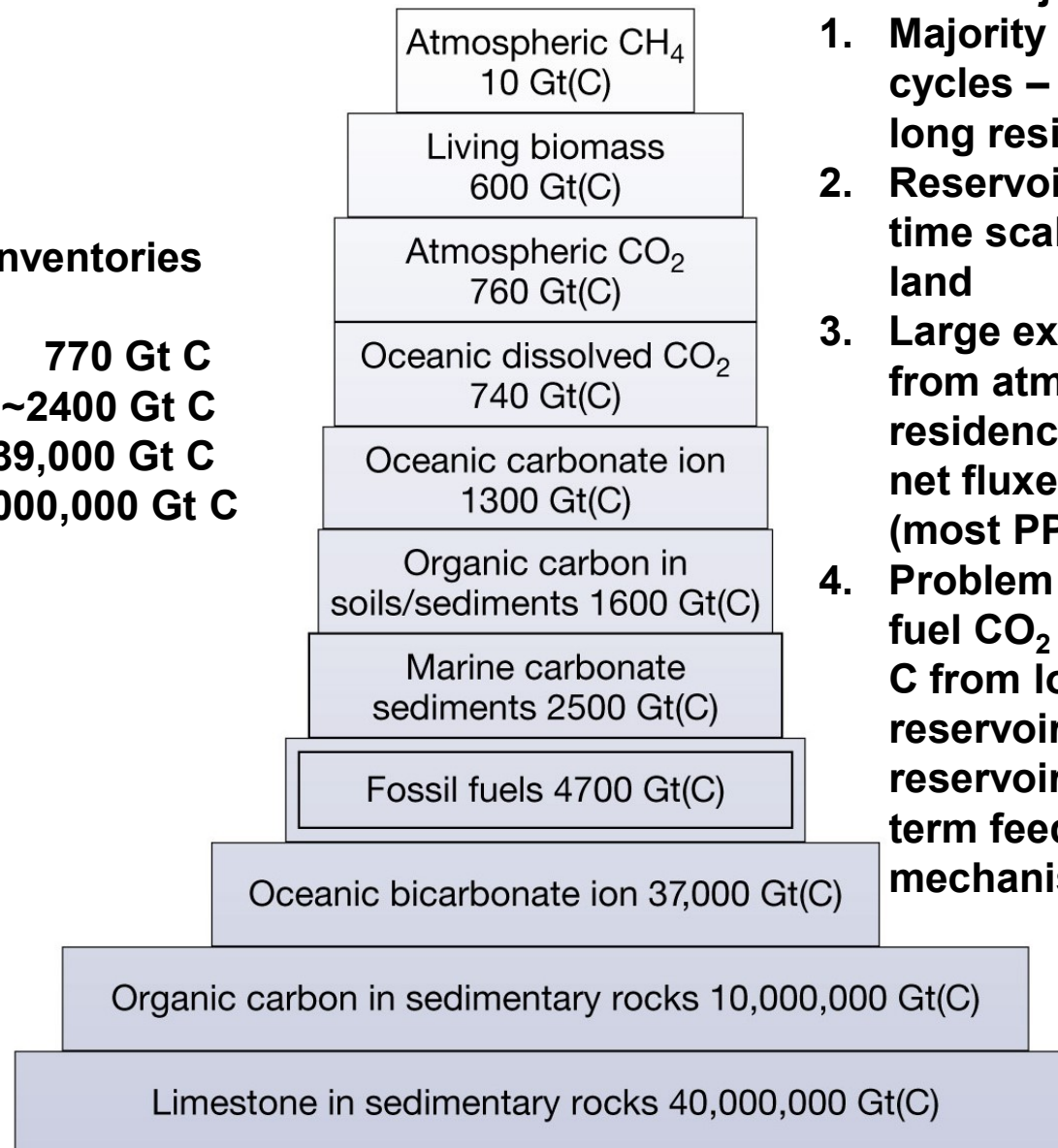
# Carbon reservoirs



# Carbon reservoirs

## Combined Inventories

|               |                 |
|---------------|-----------------|
| Atmosphere    | 770 Gt C        |
| Terr. Systems | ~2400 Gt C      |
| Oceans        | ~39,000 Gt C    |
| Sed. Rocks    | 50,000,000 Gt C |



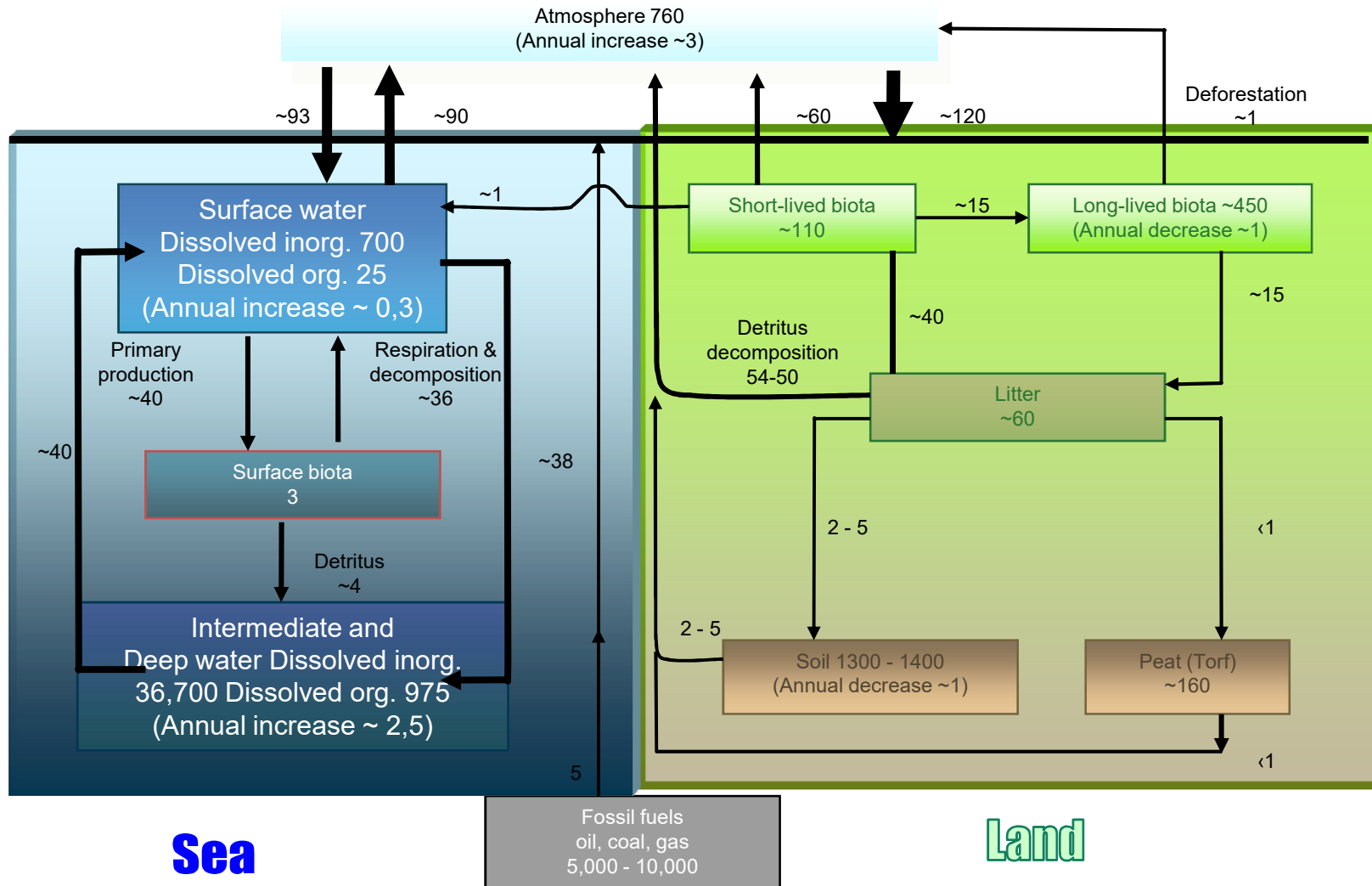
## Major Inventories

1. Majority of C tied up in rock cycles – large reservoirs with long residence times
2. Reservoirs active on short time scales are ocean, atm, & land
3. Large exchange fluxes to and from atm – atm has short residence time (3 yr); small net fluxes due to biology (most PP is respired)
4. Problem with adding fossil fuel CO<sub>2</sub> to atm – transferring C from long term geologic reservoir to a short term reservoir – may affect short term feedback control mechanisms

# The Carbon Cycle

- The carbon cycle describes the exchange of carbon atoms between various reservoirs within the earth system.
- Since our planet and its atmosphere form a closed environment, the amount of carbon in this system does not change. Where the carbon is located - in the atmosphere or on Earth - is constantly in flux.
- The carbon cycle is a geochemical cycle and since it involves the biosphere it is sometimes referred to as a bio-geochemical cycle.

# Global Carbon Cycle



Principal reservoirs and fluxes in the carbon cycle. Units are  $10^{15}$  g(Pg) C (burdens) and PgC/yr (fluxes). (From Bolin (1986))

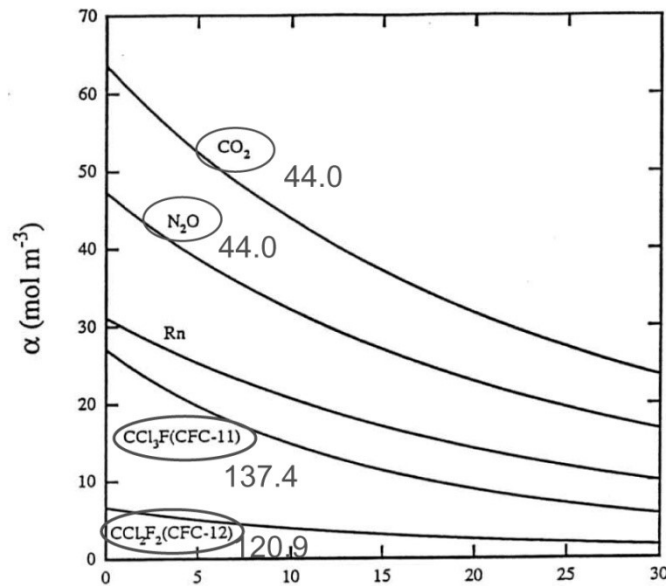
# Carbon is released into the atmosphere in several ways

- Respiration by plants and animals.
- Decay of animal and plant matter.
- Combustion of organic material
- Production of cement.
- The ocean releases CO<sub>2</sub> into the atmosphere.
- Volcanic eruptions and metamorphism

# Carbon is taken from the atmosphere in several ways

- Photosynthesis.
- The oceans when the seawater becomes cooler, more  $\text{CO}_2$  dissolve and become carbonic acid.
- In the upper ocean areas organisms convert reduced carbon to tissues, or carbonates.

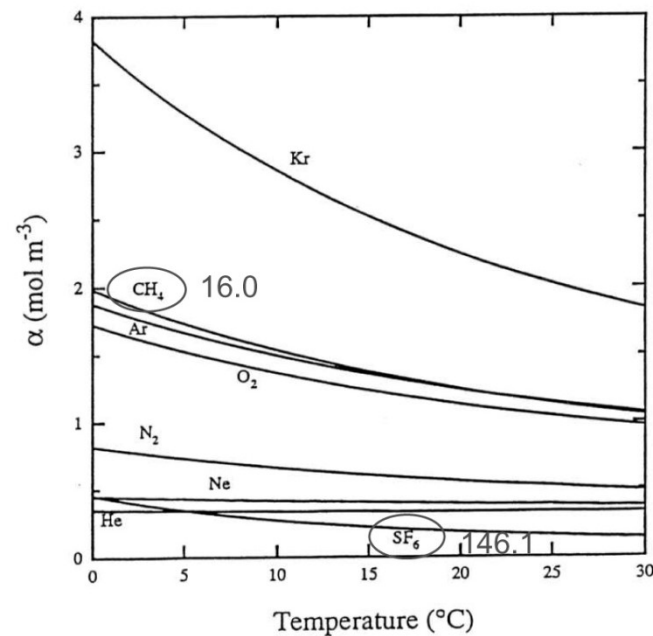
# SOLUBILITY OF GASES VERSUS TEMPERATURE



## NON-IDEAL GASES

| gas                  | molecular weight |
|----------------------|------------------|
| $\text{CO}_2$        | 44.0             |
| $\text{N}_2\text{O}$ | 44.0             |
| CFC-11               | 137.4            |
| CFC-12               | 120.9            |
| $\text{CH}_4$        | 16.0             |
| $\text{SF}_6$        | 146.1            |

↑  
Increasing solubility



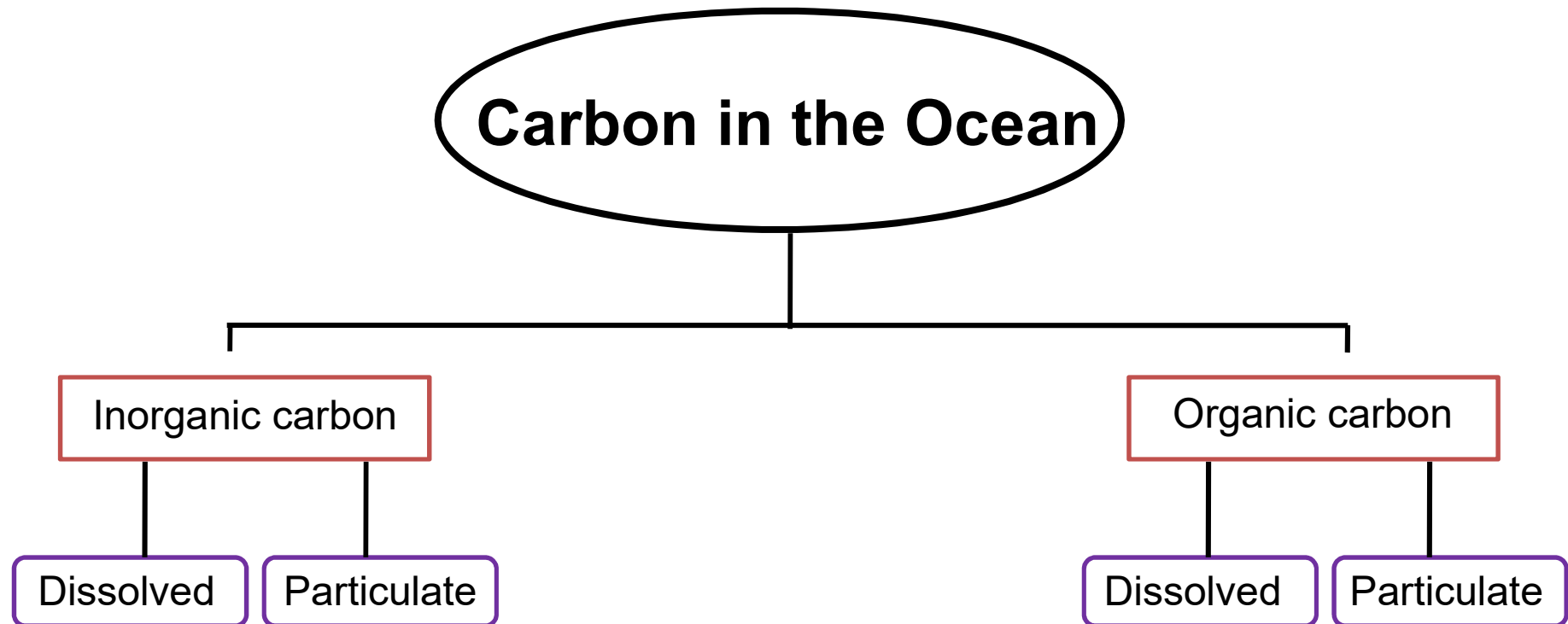
## IDEAL GASES

| gas          | molecular weight |
|--------------|------------------|
| Rn           | 222.0            |
| Kr           | 83.8             |
| Ar           | 39.9             |
| $\text{O}_2$ | 32.0             |
| $\text{N}_2$ | 28.0             |
| Ne           | 20.2             |
| He           | 4.0              |

↑  
Increasing solubility

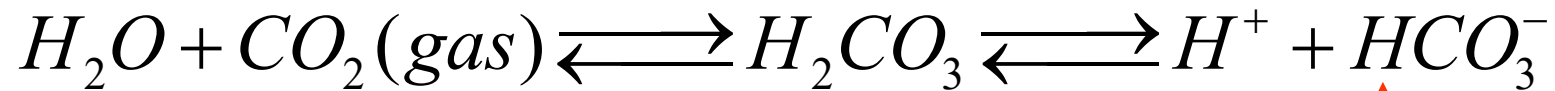
Figure 3.2.1

# Forms of Carbon in the Ocean

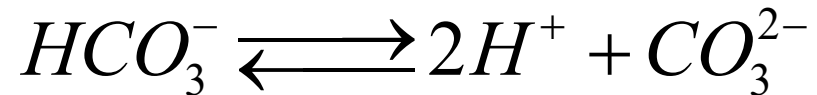


# Inorganic carbon: Dissolved

Carbonic Acid

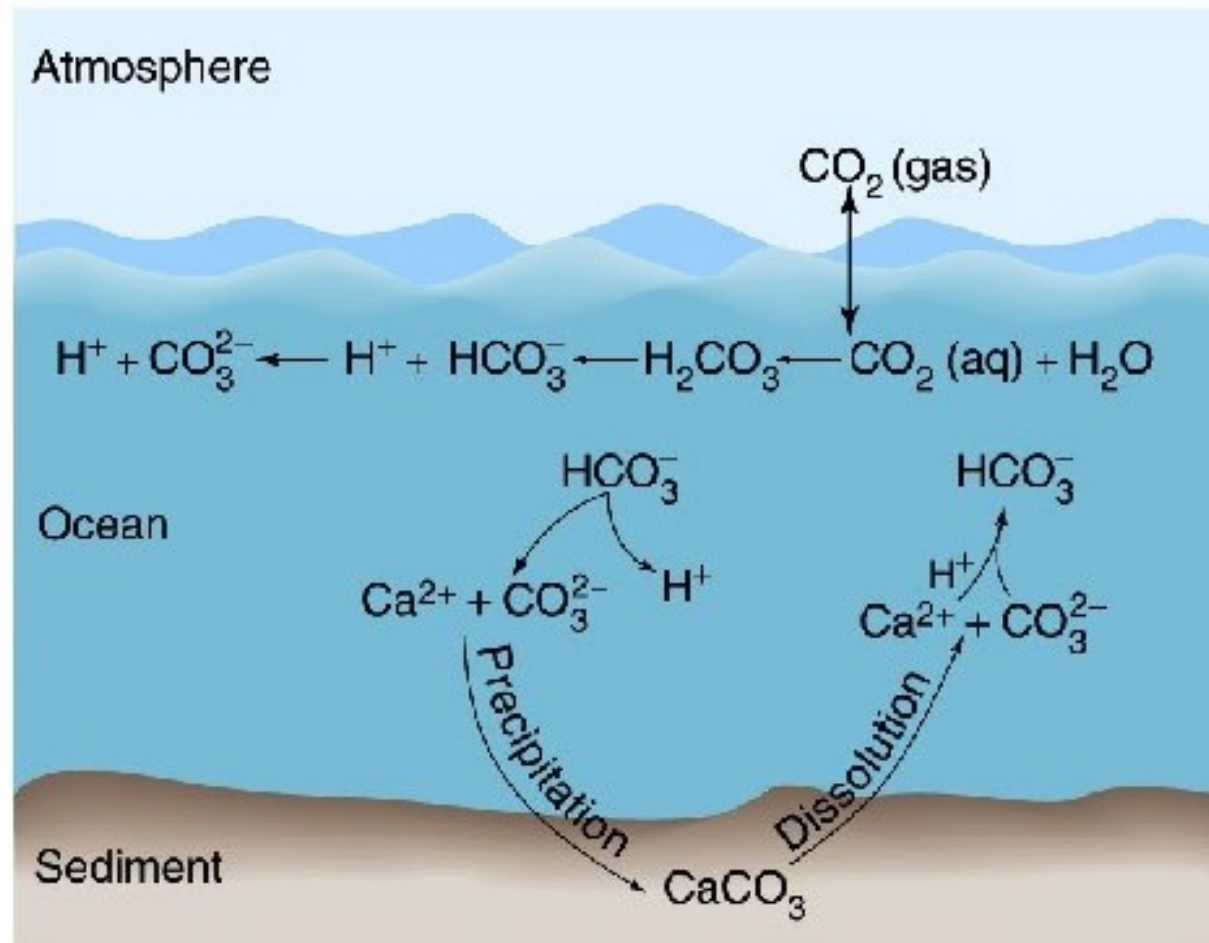
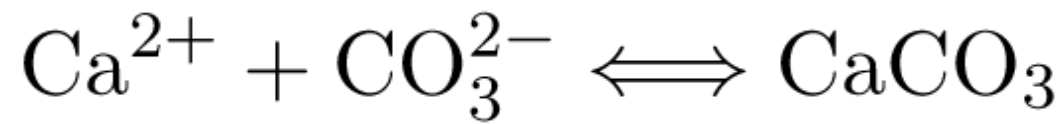


Bicarbonate Ion



Carbonate

# Inorganic carbon: Particulate



**Calcification:** Some marine organisms combine calcium with bicarbonate ions to make calcareous shells or skeletons

**CO<sub>2</sub> balance of calcification:** Calcification produces CO<sub>2</sub> !

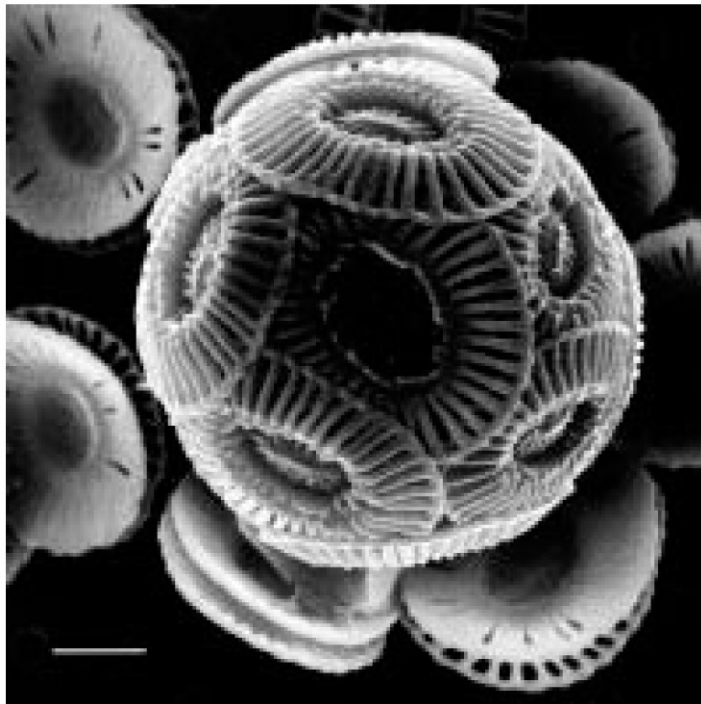


**Oceanic blooms of coccolithophorids** and production of **coral reefs**

**DO NOT** help decreasing the atmospheric increase in CO<sub>2</sub>

# Biogenic carbonate

## MARINE COCCOLITHOPHORIDS



*Emiliana huxleyi*



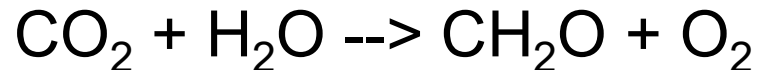
*Gephyrocapsa oceanica*

Figure 4.1.6

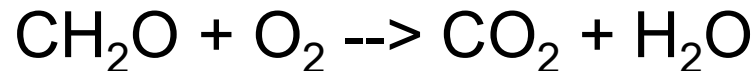
**Calcification:** calcareous shells or skeletons

# Organic carbon

The ***photosynthesis*** reaction removes carbon atoms from the atmosphere and incorporates them into the living tissue of green plants. It requires energy derived from radiation in the visible part of the electromagnetic spectrum. The chemical reaction:

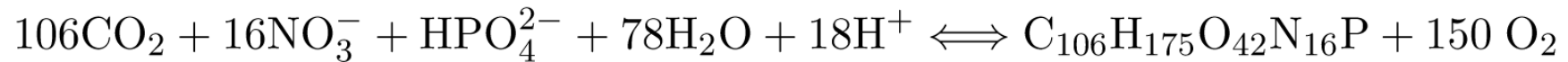


The ***respiration*** (and decay) reaction undoes the work of photosynthesis, thereby returning carbon atoms to the atmosphere:



Oxygenic photosynthesis, oxygen from sea water

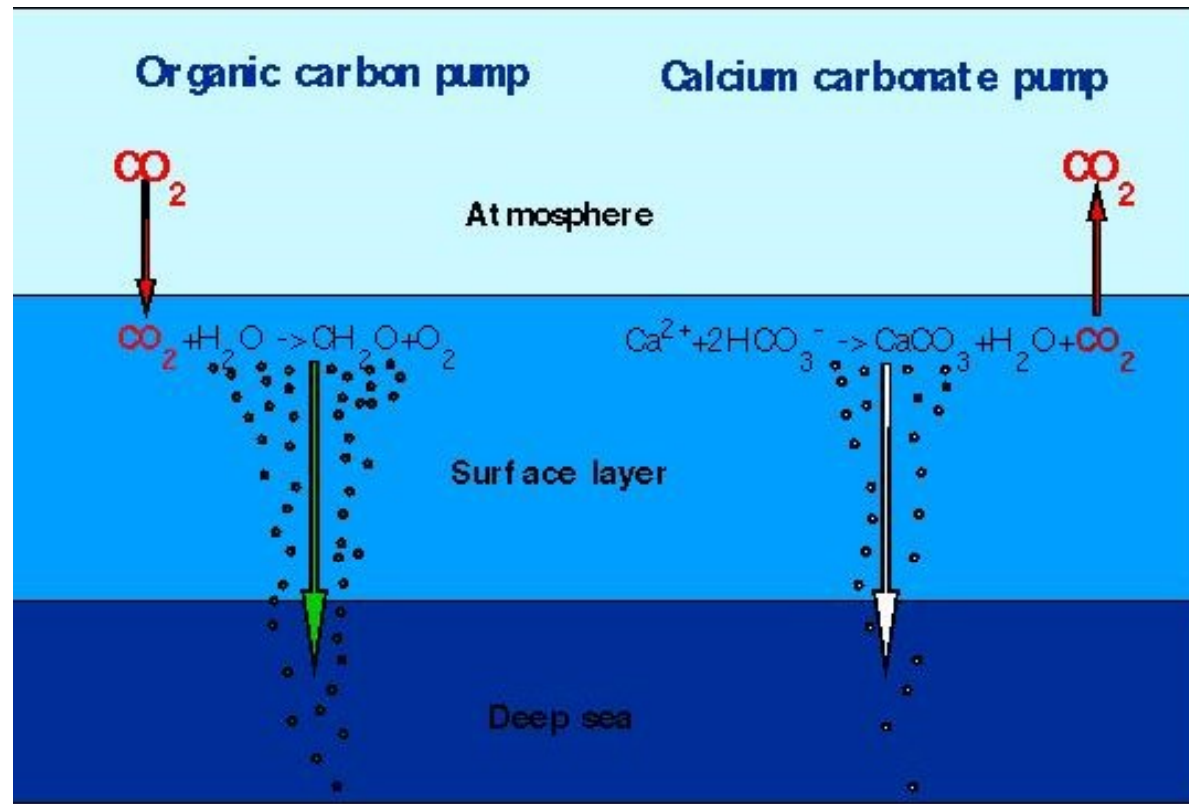
# Organic pump



The ***marine biosphere*** operates like a ***'biological pump'***. In the sunlit uppermost 100 meters of the ocean, photosynthesis serves as a source of oxygen and a sink for carbon dioxide and nutrients like phosphorous.

DIC and [H<sup>+</sup>] decrease, net consumption of CO<sub>2</sub> in the upper layers, has to be balanced by inorganic carbon by “transport sink” of CO<sub>2</sub>

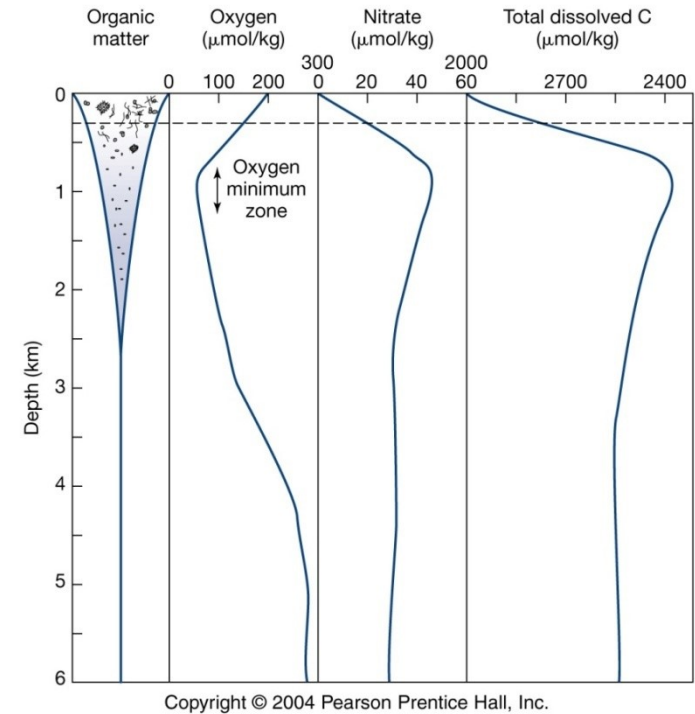
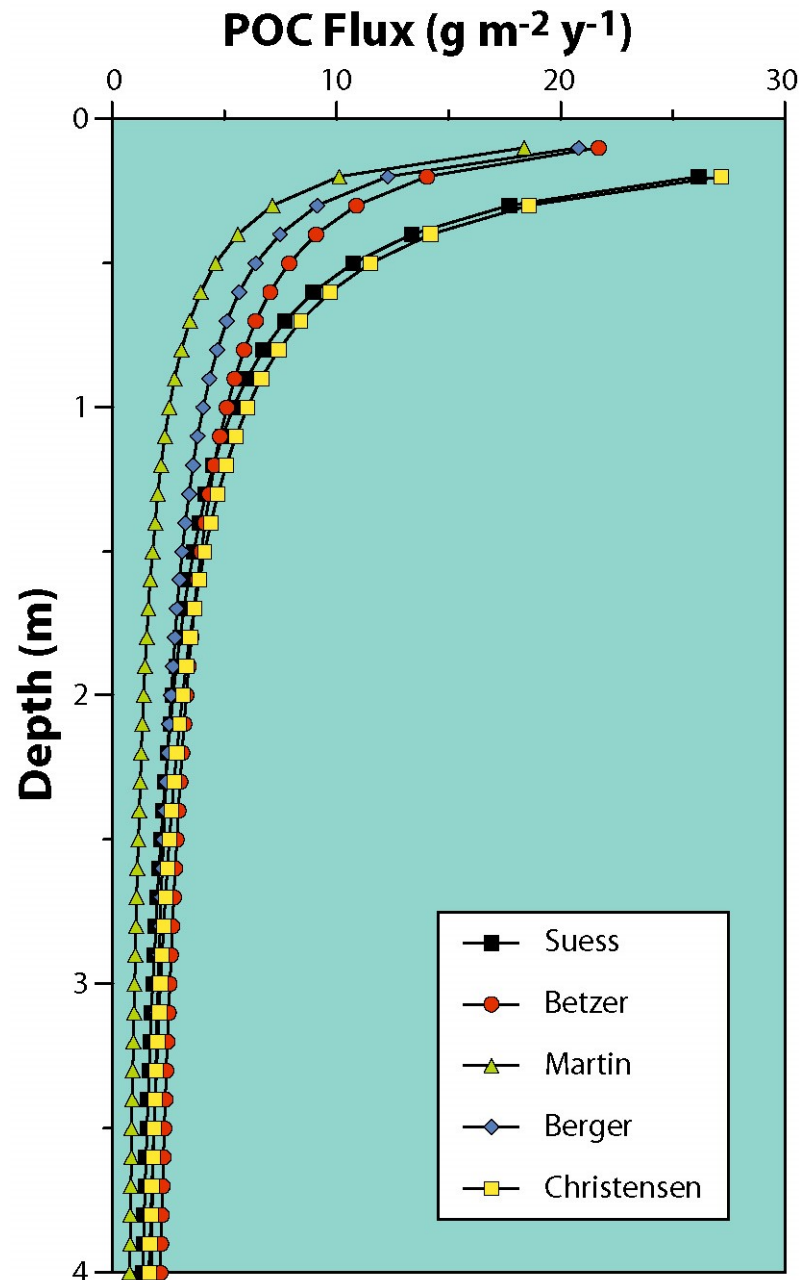
# Organic versus inorganic carbon pump



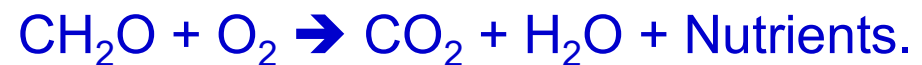
Photosynthetic carbon fixation and the flux of organic matter to depth, termed **organic carbon pump**, generates a  $\text{CO}_2$  sink in the ocean.

In contrast, calcium carbonate production and its transport to depth, referred to as the **carbonate pump**, releases  $\text{CO}_2$  in the surface layer.

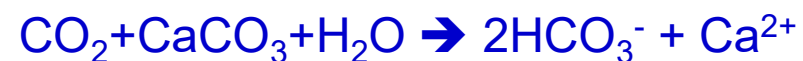
# Sinking carbon fluxes

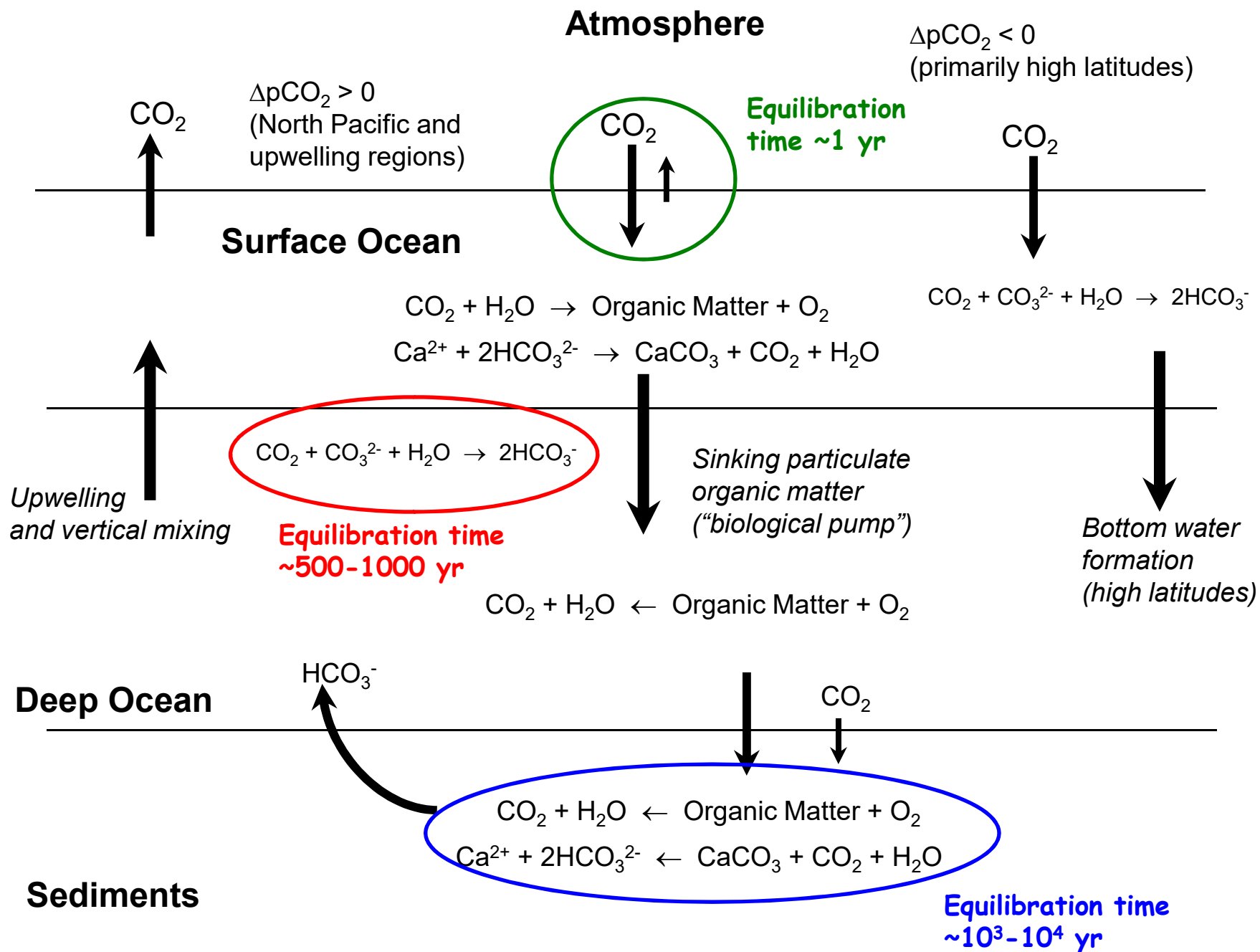


Decomposition of sinking organic carbon



Dissolution of sinking inorganic carbon

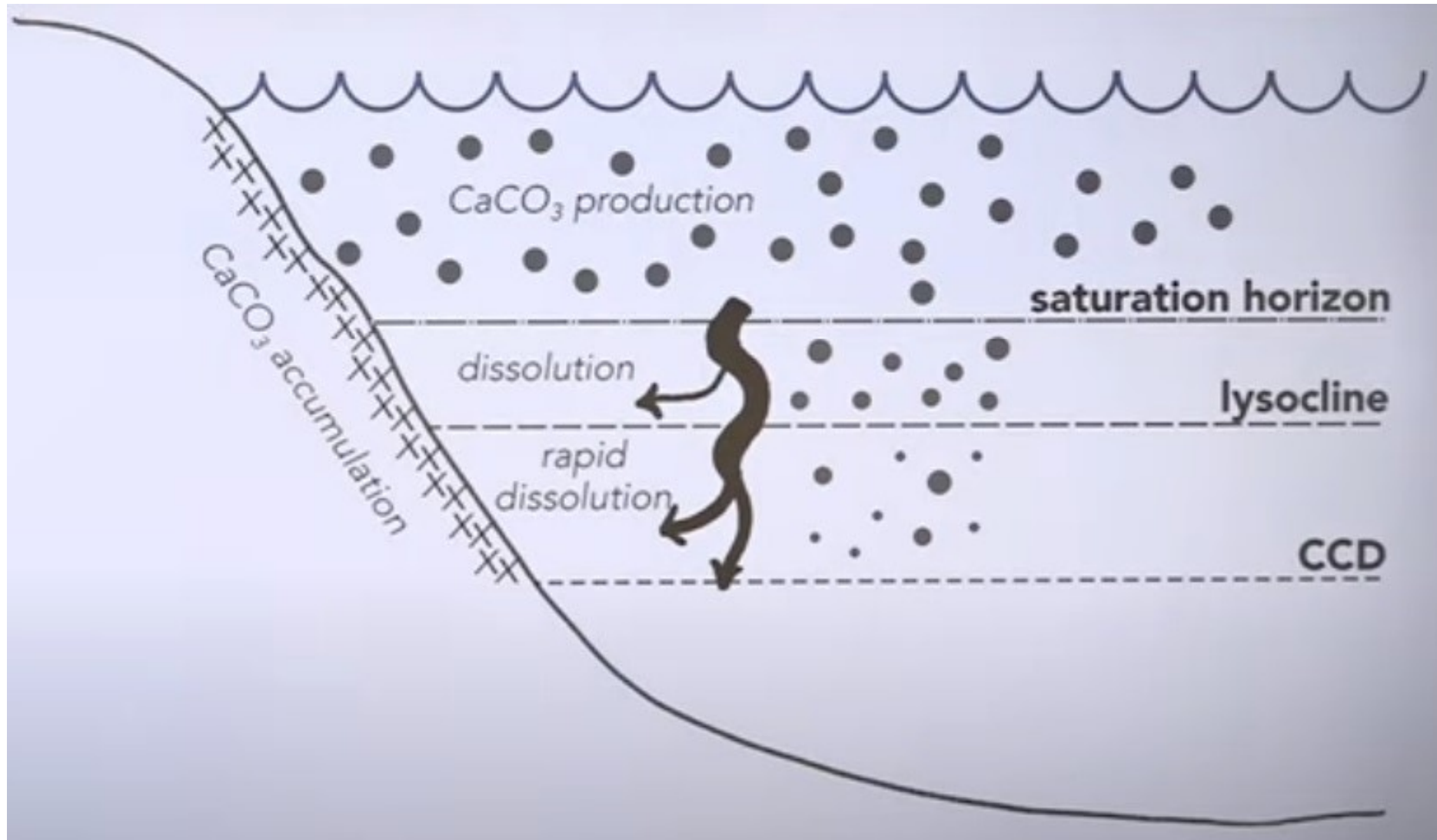


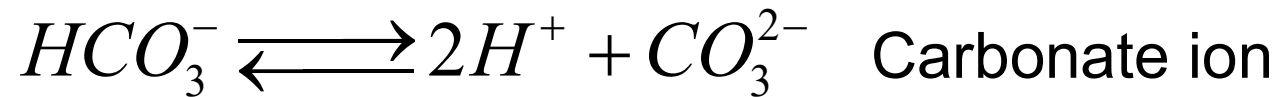
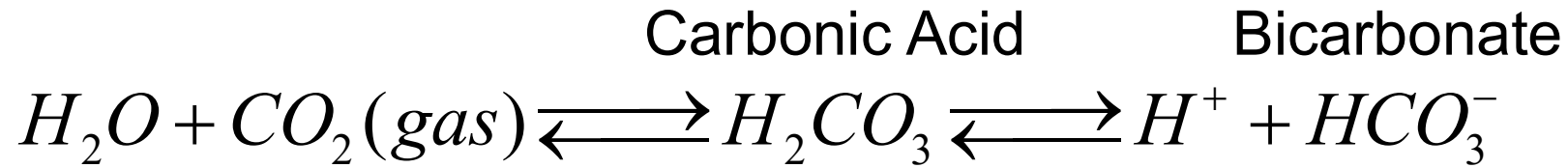


# Ocean acidification

- It is a process that occurs when carbon dioxide ( $\text{CO}_2$ ) from the atmosphere is absorbed by seawater, leading to a decrease in the pH of the ocean.
- This change in pH makes the ocean more acidic over time. The primary driver of ocean acidification is the increasing levels of  $\text{CO}_2$  in the atmosphere, mainly due to human activities such as burning fossil fuels and deforestation.
- When  $\text{CO}_2$  is absorbed by seawater, it reacts with water to form carbonic acid ( $\text{H}_2\text{CO}_3$ ). Carbonic acid then dissociates into hydrogen ions ( $\text{H}^+$ ) and bicarbonate ions ( $\text{HCO}_3^-$ ), increasing the concentration of hydrogen ions in the water and decreasing the pH.
- This process also reduces the availability of carbonate ions ( $\text{CO}_3^{2-}$ ) in seawater, which are essential for the formation of calcium carbonate ( $\text{CaCO}_3$ ) used by marine organisms such as corals, shellfish, and some plankton to build their shells and skeletons.

# Saturation Horizon, Lysocline and Carbonate Compensation Depth





**Saturation Horizon**, also known as the calcite or aragonite saturation horizon, is the depth in the ocean below which the water is undersaturated with respect to calcium carbonate minerals (calcite or aragonite). In other words, it is the depth at which the concentration of carbonate ions ( $CO_3^{2-}$ ) in the water is too low for calcium carbonate minerals to remain stable.

**Lysocline** is a depth in the ocean where there is a rapid rate of dissolution of calcium carbonate ( $\text{CaCO}_3$ ). Below the lysocline, the conditions of temperature, pressure, and acidity cause the calcium carbonate shells and skeletons of marine organisms to dissolve more readily. This depth marks the boundary between the surface layer where calcium carbonate is abundant and the deeper ocean where it is less prevalent due to dissolution.

**Carbonate Compensation Depth**. It refers to the depth in the ocean at which the rate of calcium carbonate ( $\text{CaCO}_3$ ) accumulation equals the rate of dissolution, effectively balancing the amount of  $\text{CaCO}_3$ . Below the CCD, the pressure and acidity of the water increase, making it more difficult for  $\text{CaCO}_3$  to remain stable and leading to its dissolution.

**Eutrophication** is a process in which a body of water becomes overly enriched with nutrients, leading to excessive growth of algae and other aquatic plants. This increased plant growth, known as an algal bloom, can have negative effects on the ecosystem due to the depletion of oxygen in the water when the plants die and decompose. This decomposition process consumes oxygen, leading to a decrease in oxygen levels in the water.

Eutrophication is often caused by human activities, such as agricultural runoff containing high levels of nutrients like nitrogen and phosphorus, sewage discharge, and the use of fertilizers.

**Ocean Hypoxia** refers to low oxygen levels in ocean waters, typically defined as oxygen concentrations below about 2 milligrams per liter (mg/L). Hypoxia can occur naturally in certain parts of the ocean, but human activities have significantly increased the occurrence and extent of hypoxic zones in recent years.

The main causes of ocean hypoxia include eutrophication, Climate change and ocean circulation:

Ocean hypoxia can have significant impacts on marine ecosystems. It can lead to the death of fish and other marine organisms, alter food webs, and reduce biodiversity. Hypoxic conditions can also affect nutrient cycling and the productivity of marine ecosystems.



Thank You

Pillow lava at CIR observed during manned submersible dive